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On the Role and Effectiveness of Fiscal Rules and Independent Fiscal Councils

Roel Beetsma, W. Raphael Lam, Erik Leonards, Marcos Poplawski-Ribeiro, Alexandra Solovyeva

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On the Role and Effectiveness of Fiscal Rules and Independent Fiscal Councils*

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ABSTRACT: Fiscal rules and fiscal councils are often adopted independently: many countries operate under fiscal rules without a fiscal council, while fiscal councils are frequently established only after a fiscal rule is already in place, suggesting that the two institutions may contribute differently to fiscal discipline. This paper starts with a simple theoretical framework to explain these stylized facts, accounting for the fact that governments, or even society at large, may not always be in favour of a fiscal council. The model also predicts that a sufficient strengthening of a budget balance rule, an independent fiscal council, and an increased interaction between rule strength and council strength all help reduce the expected deficit. The ensuing empirical analysis, using the updated 2025 IMF datasets on fiscal rules and fiscal councils, tests these predictions for a large country sample, including 32 advanced economies and 58 emerging market and developing economies spanning from 2000 to 2024. We find indicative evidence that, depending on the sample and the institutional design, both stronger rules and councils are associated with lower primary deficits, reduced projection biases and improved forecast accuracy. We also find some role for the interaction between the strengths of rules and councils, although the pattern varies.

Keywords:	Fiscal rules, fiscal councils, IMF databases, fiscal outcomes, fiscal projections, forecasting accuracy
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I. Introduction

The gradual increase in fiscal rules and independent fiscal councils over the past three to four decades calls for a comprehensive evaluation of their effectiveness in promoting fiscally sustainable public finances by reducing budget deficits, reducing fiscal forecast errors, and improving fiscal forecast accuracy.[1] This paper offers a first empirical analysis of these questions using a large sample of 90 countries, including 32 advanced economies (AEs) and 58 emerging market and developing economies (EMDEs), spanning from 2000 to 2024, based on the IMF's comprehensive and recently updated dataset on fiscal rules and councils (Alonso et al., 2025). These authors define fiscal rules as "long-lasting constraints on fiscal policy through numerical limits on broad budget aggregates, such as public debt, the fiscal balance, government expenditures, or revenues" and fiscal councils as "independent nonpartisan agencies with a mandate to provide fiscal oversight". Fiscal councils take many institutional forms, ranging from stand-alone entities to budgetary watchdogs such as parliamentary committees to general audit offices. Independent fiscal councils help to enhance fiscal transparency (Debrun et al., 2013, Beetsma et al., 2022, and Acalin et al., 2025) and accountability, thereby promoting long-run fiscal sustainability. Their main activities differ, but the most common ones are assessing or producing macro-fiscal forecasts, monitoring compliance with fiscal rules, evaluating public debt sustainability, and providing recommendations on budgetary policies (Beetsma, 2023). Fiscal councils often have no formal decision powers over fiscal policy, but they may fulfill an important role in the budgetary framework, for example by providing technical analysis on macroeconomic and budgetary positions or assessing the government's macroeconomic or even budgetary forecasts for the preparation of national budgets.

The EU has been one of the leading blocs in the development of fiscal rules and fiscal councils, with the introduction of its Stability and Growth Pact (SGP) in the late 1990s. The SGP has featured 3 and 60 percent of GDP reference values for fiscal deficits and public debt, respectively. Next to the SGP, countries in the EU have also introduced national fiscal rules—see Debrun et al. (2008) for an early assessment. In 2011, the EU adopted a Directive on budgetary frameworks that has been revised recently along with the 2024 reform of the EU economic-governance framework. The Directive envisages a prominent role for independent fiscal councils. Almost all EU countries now have a fiscal council.

The spread of fiscal rules and fiscal councils has extended well beyond the EU. We observe a general trend of adoption of both, as well as a gradual strengthening of the former. By the end of 2024, there were about 120 countries with at least one fiscal rule and more than 50 with a fiscal council (Alonso et al., 2025). While advanced economies have been in the lead in adopting rules and councils, others are following suit. Further, the establishment of a fiscal council, if it happens, tends to lag behind the adoption of fiscal rules, which suggests a relationship between the introduction of a fiscal council and the presence of fiscal rules. Fiscal councils may play a role complementary to fiscal rules in strengthening fiscal discipline when rules alone may not be sufficient.

Countries have adopted fiscal rules as guardrails to foster fiscal discipline. Empirical evidence suggests that fiscal rules could foster budgetary discipline (Azzimonti et al., 2016), signal a government's commitment to sound

[1] Forecast accuracy is measured by the absolute size deviation between realization and initial forecast.

fiscal management (Hatchondo et al., 2022a, 2022b), mitigate procyclical spending (Reuter et al., 2022; Eyraud et al., 2023), and coalesce political debate (Cao et al., 2024). The IMF Fiscal Rules and Fiscal Council databases show that many countries with fiscal rules still do not have a council, and a fiscal council is often introduced once a fiscal rule is already in place, suggesting rules and councils could contribute differently to fiscal discipline. While the existing literature has examined the impact of fiscal rules or fiscal councils on fiscal outcomes, including the interactions between fiscal rules and councils (e.g., see Debrun and Kumar (2008), Maltritz and Wüste (2015) and Caselli and Reynaud (2020)), we present new evidence using recent 2025 data, based in particular on the strengths (in the sense that they contain desirable design features) of rules and councils, and how these interact, in affecting primary deficits, their forecasting errors and their forecasting accuracy.

To better understand the relationship, we present a simple theoretical framework that can account for the fact that fiscal rules may exist in the absence of a transparency-enhancing fiscal council because the government, or even society, prefers not to have such a council. For the government, the resulting increase in fiscal transparency makes it more difficult to disguise fiscal profligacy as the result of adverse economic developments. Society, in turn, may not want a fiscal council because of adverse incentive effects impacting the deficit. We argue, however, that the more relevant equilibrium is when a fiscal council is implemented that strengthens the incentive for fiscal discipline. The framework also provides a number of hypotheses that are empirically tested in the paper. In particular, the model predicts that a sufficient strengthening of the rule as well as of the fiscal council, and an increased interaction between the two, all reduce the expected deficit.

Those predictions are examined by employing an econometric analysis with data from the updated IMF databases on fiscal rules and councils (Alonso et al., 2025). Those databases allow us to construct an unbalanced panel for 90 countries—32 advanced economies (AEs) and 58 emerging market and developing economies (EMDEs)—spanning from 2000 to 2024. With this panel, we first estimate the determinants of the primary deficit and test whether the presence of a stronger fiscal rule, a stronger fiscal council, or a combination of both is associated with a lower primary deficit. We then explore the determinants of forecast errors in primary deficits, the accuracy of these projections, and whether stronger fiscal rules, councils, or both, reduce those projection biases and improve their accuracy.

Our main findings are the following. In line with the hypotheses outlined above, both stronger fiscal councils and stronger fiscal rules tend to be associated with lower primary deficits, reduced forecast errors, and improved forecasting accuracy. In terms of economic significance, our estimates suggest that shifting from the 25th to the 75th percentile of the distribution of the fiscal rule strength would be associated on average with a reduction in the primary deficit of 1.2 percentage points of GDP. For fiscal council strength, the same shift would be associated with a reduction in the primary deficit of 0.4 percentage point of GDP.

The results also suggest that the interaction between rule strength and council strength matters for forecasting errors of primary deficits and their accuracy, even though a consistent pattern is not fully obtained, most likely because of the limited number of observations on councils. Moreover, it is important to bear in mind that fiscal achievements are driven not only by rules and councils but by other factors, such as the fiscal discipline culture and procedures governing the budgetary process, as well.

This paper relates to several strands of the literature. First, there is a large literature on the motivation and implementation of fiscal rules. Examples include Beetsma and Uhlig (1999), Chari and Kehoe (2007, 2008), and Davoodi et al. (2022a). Esquivel et al. (2026) demonstrate that the optimal debt limit depends on the nature of spending needs. The limit should be higher with productivity-enhancing spending and tighter in the presence of wasteful spending. Empirical analyses on the effectiveness of fiscal rules have been conducted by, for example, Reuter (2015) and Caselli and Wingender (2021), who find that EU countries try to steer fiscal outcomes toward the 3 percent-of-GDP reference value for the deficit, thereby effectively improving their fiscal position, even if they do not fully observe the rule; Asatryan et al. (2018), who demonstrate a fall in the debt-to-GDP ratio associated with a constitutional-level budget balance rule; Reuter (2018), who explores the determinants of compliance with national numerical fiscal rules; the European Fiscal Board (2019), which provides an assessment of the compliance with the headline numerical values of the various rules in the SGP; Kraemer and Lehtimaki (2023), who provide evidence to suggest that both the EU and national fiscal rules yield lower public debt ratios; and Eklou and Joanis (2020), who find that developing countries with fiscal rules in place tend to have a more muted political budget cycle in public consumption during election years, with the effectiveness of these rules hinging on their institutional design and on the degree of competitiveness of the election. Protrafke (2025) provides an overview of the empirical literature on the economic consequences of fiscal rules. We contribute to this literature by presenting a simple theoretical framework that accounts for the effects of (stronger) fiscal rules and exploring their role using the most up-to-date IMF database of fiscal rules.

Second, there is a literature on fiscal councils and their role in improving fiscal transparency (Beetsma et al., 2022; and Sloof et al., 2026). Third, there is a significant literature on fiscal transparency not specifically linked to fiscal councils. For example, see Bordignon and Minelli (2001), Milesi-Ferretti (2004), Alt and Lassen (2006), Alt et al. (2006), Shi and Svensson (2006), Besley and Smart (2007), Beetsma et al. (2022), Dovis and Kirpalani (2021), Lledó and Poplawski-Ribeiro (2013), and Lopez-Murphy et al. (2026), the latter two for low-income countries (LICs). We contribute to these strands of the literature by developing a simple theoretical framework which also highlights how enhanced fiscal transparency affects fiscal outcomes, and testing this using the most up-to-date IMF database of fiscal councils, based on the assumption that (stronger) fiscal councils enhance fiscal transparency.

The remainder of the paper is organized as follows. Section II describes the data. We summarize the most relevant elements of the updated IMF databases on fiscal rules and fiscal councils. Section III presents the theoretical framework. Section IV presents and discusses the results of our empirical analysis on primary deficits, while Section V does this for their forecasting errors and the accuracy of the latter. Finally, Section VI concludes the main text.

II. Data Sample and Coverage

The paper uses a large sample of 90 countries across AEs (32) and EMDEs (58). The full list of countries and the descriptive statistics of all variables used in the analysis can be found in Annex A. Our sample period runs from 2000 (in most cases) to 2024, depending on data availability.

II.1. Latest IMF Fiscal Rules and Fiscal Council Databases (1985-2024)

The paper uses the latest 2025 versions of the IMF Fiscal Rules and Fiscal Councils databases, which contain detailed information on fiscal rules and fiscal councils in over 120 countries. However, data limitations prevent the inclusion of the full sample (see Annex Table A.II). Our narrower focus implies that some countries without reliable estimates of fiscal statistics in earlier periods are excluded. A few countries in Africa with only supranational rules but not national rules are also excluded.

The number of countries with fiscal rules has more than doubled since the beginning of the century. As of end-2024, 122 economies have adopted at least one numerical fiscal rule (Alonso et al., 2025). The first group of economies to adopt fiscal rules were the advanced economies, with adoption in the early 1990s. The emerging market economies followed with adoptions mostly taking off since the mid-1990s. By the late 1990s, fiscal rule frameworks were already established in most AEs, largely due to the debt and deficit criteria set out by the Maastricht Treaty in the European Union (EU). In EMDEs, national fiscal rules were often initially adopted to commit to fiscal adjustments after crises, to consolidate gains from reforms, or to prevent procyclical spending caused by volatile natural resource prices and save for future generations.

In terms of design, the most common rules set limits on the budget balance, closely followed by debt rules, while revenue rules are less common. Many countries have adopted multiple rules at the same time to constrain different fiscal aggregates. The most common is a combination of a budget balance rule and a debt rule, while budget balance rules and debt rules are deployed as stand-alone rules only in a minority of cases (Alonso et al., 2025).

Latest data show that fiscal councils are mostly established among AEs (IMF, 2025a; 2025c; and Alonso et al., 2025). Panel I of Figure 1 depicts the number of countries with an independent fiscal council (IFC), with a few countries having more than one fiscal oversight institution. This is the case for some member states of the EU, such as Belgium and the Netherlands, where the roles mandated under the EU Directive on budgetary frameworks may be split between two IFCs.[2] For example, the Netherlands Bureau of Economic Policy Analysis constructs the official macroeconomic projections, while the Dutch Council of State monitors the adherence to the national fiscal rules, as mandated by the Directive, and to the EU fiscal rules.

The initiative to establish fiscal councils varies across countries, with either the legislative or executive branches of government taking the lead. In some cases, fiscal councils arise from supranational legislation, as observed in EU member states.[3] In the United States, the legislative branch created the Congressional Budget Office (CBO) through the Congressional Budget and Impoundment Control Act of 1974, which was subsequently signed into law by President Richard Nixon. In Brazil, although the executive branch proposed a law adopting fiscal rules in 2000 that included the possibility of a fiscal council, it was the Senate that formally advocated for its creation 15 years later by approving a Senate proposal from the previous year. In many other countries, however, the executive branch has been the primary proponent of establishing fiscal councils. This is the case in South Africa,

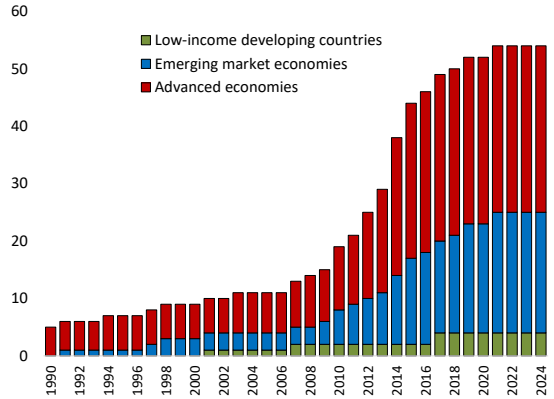
[2] Precisely, the Council Directive 2011/85/EU of 8 November 2011 on requirements for budgetary frameworks of the Member States, see https://eur-lex.europa.eu/eli/dir/2011/85/oj/eng.

[3] In some instances, a fiscal council was already present before EU supranational legislation mandated them.

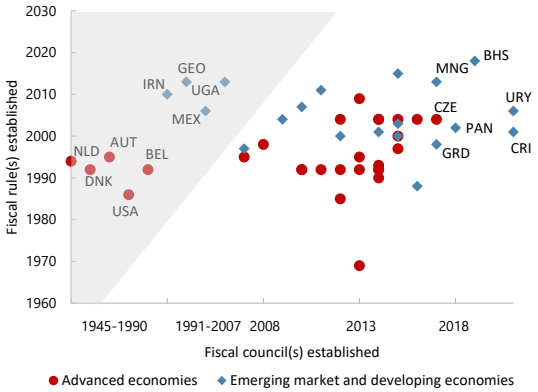
for example, where the Financial and Fiscal Commission (FFC) was established in 1993 by the executive branch to support fiscal discipline and intergovernmental fiscal relations.[4] As we discuss in the next section, these differences in who initiates the creation of fiscal councils may reflect varying levels of commitment to fiscal transparency and illustrate potential trade-offs faced by the executive branch.

Figure 1. Countries with Fiscal Councils and Combination with Fiscal Rules

I. Fiscal Councils
(Number of councils by country groups)



II. Establishment of Fiscal Rules and Fiscal Councils
(Year)



Sources: IMF Fiscal Councils Database (2024); and Alonso et al. (2025).
Note: The right-hand figure shows years of establishment of fiscal councils and fiscal rules for countries that have both fiscal councils and one or more fiscal rules as of end-2024. The shaded area corresponds to countries where fiscal councils were established earlier than the first fiscal rule.

An interesting observation is that almost all countries with a fiscal council have adopted numerical fiscal rules, while the opposite, countries with a fiscal council but no fiscal rules, is almost nonexistent; Korea is the sole exception (Table 1). The data also show that the introduction of a fiscal council has generally trailed behind the adoption of numerical fiscal rules. Panel II of Figure 1 shows that for all countries with at least one fiscal rule and a fiscal council in 2024, the years in which the first rule and the fiscal council were established. In most cases, the introduction of a fiscal rule preceded that of a fiscal council. That the adoption of fiscal rules tends to precede the fiscal council is partly because fiscal councils are often introduced to strengthen the fiscal rules framework by increasing fiscal transparency (as will be illustrated by our theoretical framework next) and by monitoring the implementation of the rules, suggesting a potential key complementarity between fiscal rules and fiscal councils in fostering fiscal discipline.

[4] See https://www.ffc.co.za/faq.

Table 1. Number of Fiscal Rules and Fiscal Councils by Income Group, 2024

	Advanced Economies	Emerging Markets and Developing Economies	Total
Both fiscal rule(s) and fiscal council(s)	27	20	47
Only fiscal rule(s)	8	65	73
Only fiscal council(s)	1	0	1

Sources: IMF Fiscal Rules Dataset, 1985–2024; IMF Fiscal Councils Database, 2024; and Alonso et al. (2025).

Key to the effectiveness of fiscal rules and fiscal councils is their design. As the design covers different features and attributes of the fiscal rules, the paper adopts the "strength" index measures of fiscal rules and fiscal councils as prepared in Alonso et al. (2025), who describe in detail the construction of a strength index for rules and fiscal councils (Annex A in Alonso et al. 2025). The methodology creates a composite index to measure how well fiscal rules in countries stand relative to the desirable design properties such as flexibility in the face of severe shocks, coverage of the rules on fiscal aggregates, statutory basis of fiscal rules, compliance, and the corrective mechanism in case rules are breached. Specifically, the strength of a fiscal rule is based on the following institutional criteria, namely (i) the statutory basis of the rule; (ii) monitoring of the rule; (iii) enforcement and correction mechanism; and (iv) flexibility and resilience against shocks. The scores on the different criteria are summed to a composite index with appropriate weights across different types of rules and coverage (with the general government receiving a weight of one and the central government of 0.75). In the case of multiple rules, scores of these rules are summed, with each additional rule getting a declining weight from the previous one. The result is standardized using the unconditional mean and standard deviation.

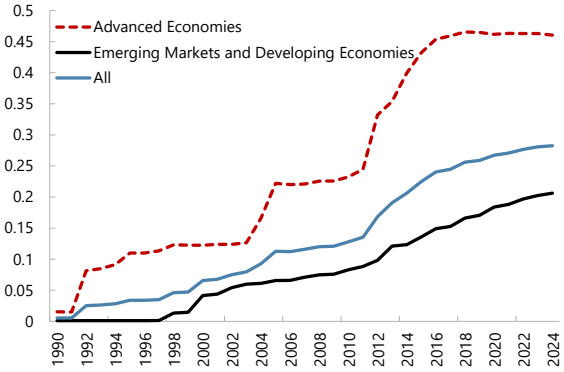
Figure 2, Panel I, depicts the evolution over time of the strength of the national fiscal rules for each country group. Strength has on average improved over time both for AEs and EMDEs.

On the fiscal councils, the paper also includes the indicators used in Alonso et al. (2025) to quantify how individual fiscal councils compare with desirable properties highlighted in IMF (2013) and the Organisation for Economic Co-operation and Development's (OECD) key principles of effective fiscal councils. Fiscal council strength is measured based on three criteria: (i) key mandates and tasks performed; (ii) operational independence and accountability; and (iii) communication channels and influence. The overall index is calculated by summing up individual scores of each criterion, each ranging between 0 and 1. Panel II of Figure 2 shows that the calculated index exhibits a large variation across fiscal councils when benchmarked against some desirable principles, with higher average scores in advanced economies, particularly in Europe.

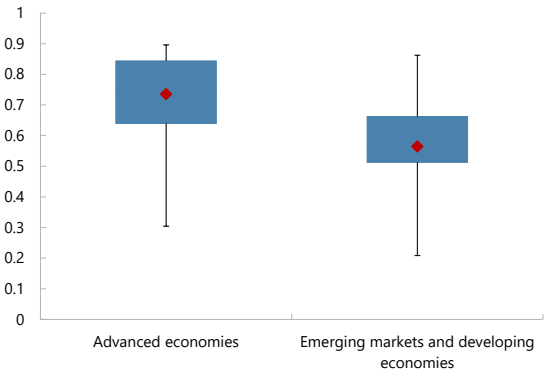
The calculated index on fiscal councils shows a positive and significant correlation (coefficient of 0.73) with the strength index on fiscal rules across countries, suggesting the complementarity of strong fiscal rules and fiscal councils.

Figure 2. Strength of Fiscal Rules and Fiscal Councils

I. Fiscal Rules Strength Index



II. Fiscal Council Strength Index across Income Groups, 2024



Sources: IMF Fiscal Rules Dataset, 1985-2024; IMF Fiscal Councils Database, 2024; and Alonso et al. (2025).
Note: The left-hand figure shows the average value of fiscal rules strength indices for each country group. The right figure shows the distribution of how fiscal councils across countries stand relative to desirable design properties in accordance with OECD recommendations (OECD 2014). The red dot shows the median level, while the interquartile range is shown in a blue box, with the minimum and maximum values shown by the whiskers.

II.2. Other Macroeconomic and Fiscal Data

The paper also uses other macro-fiscal data from additional sources. It takes macro-fiscal data from the IMF World Economic Outlook, including the inflation rate, real GDP growth, and their medium-term projections. Fiscal deficits and government debt are also retrieved from the IMF's World Economic Outlook database. Annex A (Table A.I) lists the specific definitions of these variables.

To limit an unduly large influence of outliers on the outcomes of the empirical analysis, we winsorize the primary deficit data by setting the 1 percent largest and 1 percent smallest values of the primary deficit as a percentage of GDP and their forecasts to their respective 99th and 1st percentiles.[5] Because the data for the AEs and EMDEs exhibit systematically different variances, we winsorize the data separately for AEs and EMDEs, merging the data ex post for the estimations with the full country sample. Annex A (Table A.III) presents the descriptive statistics of the sampled data. Given that some observations for a small number of countries and years are missing, we linearly interpolate these from the observations of the year before and the year after.[6]

[5] We do not winsorize the inflation data. Doing so would have no effect on the results.

[6] The interpolation is done for around 10 observations and is only potentially relevant in the regressions for the forecasting errors and accuracy in Section V, while it has no effect on the estimates for the primary deficits in Section IV.

III. A Simple Theoretical Framework on the Interactions between Fiscal Rules and Councils

Section II presented some stylized facts, some of which we highlight here. First, not all countries decide to adopt fiscal rules and even fewer establish a fiscal council. Second, while some countries with fiscal rules choose to establish a fiscal council, others do not do so. Third, when a fiscal council is established, usually one or more fiscal rules are already in place, while the opposite is much less common, which may indicate that fiscal councils tend to be established in order to support the functioning of fiscal rules. This section presents a simple illustrative theoretical framework that helps to rationalize the second fact in particular and also sheds some light on the other two facts. Other stylized facts highlighted in Section II will not be explicitly addressed.

Fiscal councils can be established at the initiative of the legislature or the government (or resulting from supranational legislation, as was the case for EU countries). Governments may not always be enthusiastic about independent fiscal oversight, because such oversight may provide different views on the state of the economy and whether the government complies with the fiscal rules. In other words, the fiscal council may help reduce the asymmetric information between the public and the government. The presence of a well-designed fiscal council could make it easier for voters or the legislature to conclude whether a transgression of the rules was justified by circumstances outside the government's control or not. Hence, if there is a political cost of noncompliance with fiscal rules, the fiscal council may be complementary to fiscal rules in inducing fiscal discipline. However, there may be instances in which a government does have an incentive to install a council, for example to assuage the financial markets in order to keep borrowing costs in check or maintain certain credit ratings, or because of its desire to constrain future administrations. Also, which entity takes the initiative to establish a fiscal council may depend on where the fiscal powers are primarily located. Overall, however, one would expect society at large to have a stronger interest in establishing a (sufficiently strong) council than the government.

We now present a simple framework that confirms this intuition. However, we show that there are also configurations in which neither the government nor society wants a (stronger) council. The framework also allows us to formulate several hypotheses that can be put to a formal empirical test in the next section. The framework takes inspiration from a more elaborate literature on transparency in the context of monetary policy. For example, Atkeson et al. (2007) show that in the absence of commitment, transparency (which is higher when monetary policy takes the form of setting the interest rate, a price), allows private agents to detect deviations from the central bank's intended or announced policy and "punish" it accordingly, thereby deterring the central bank from deviating from its plan. In their context, transparency is highest when the central bank sets the interest rate, a price. In Kostadinov and Roldan (2025), transparency is captured by the precision with which the private sector observes the chosen policy. Full transparency is optimal in their setting. That is not necessarily always the case. In Dovis and Kirpalani (2021) the optimal degree of transparency depends on the initial reputation of the policymaker, which is preserved in the presence of lower transparency. In our context, transparency is the probability that the signal about the state of the economy is accurate.

The model is deliberately kept as simple as possible. It features two periods, 1 and 2. Society's preferences (represented by the legislature) are quasi-linear and given by:

$$V^S = u(g_1) + g_2, \quad (1)$$

where g_1 and g_2 are government spending in periods 1 and 2. The function $u(.)$ is strictly increasing and strictly concave. For convenience we set society's discount factor to 1. The incumbent government faces an election at the end of period 1 in which it is re-elected with exogenous probability $0 < \pi < 1$.[7] It only obtains utility from government spending that is done by itself (on causes closest to its ideological heart). Its preferences are:

$$V^G = u(g_1) + \pi[g_2 - E(c)], \quad (2)$$

where $E(c)$ is the expected cost associated with being caught in an unwarranted (i.e., not justified by exogenous economic circumstances) transgression of a fiscal rule that takes the form of a numerical reference value $\bar{d}$ for the deficit. The cost c can be thought of as a reputational cost or a financial cost, for example because of a repricing of government debt. A higher cost c effectively amounts to a *stronger* fiscal rule. The cost is an increasing function of the strength of the fiscal rule. The limiting case of $c = 0$ corresponds to the absence of a rule or a completely ineffective rule.

The government's period budget constraints are

$$g_1 = y_1 + d, \quad (3)$$
$$g_2 = \bar{y} - d, \quad (4)$$

where y_1 is period 1 income which can take two values, y_L with probability λ and y_H with probability $1 - \lambda$, where $y_L < y_H$. Income in period 2 is non-stochastic at $\bar{y}$. Variable d is the fiscal deficit, which in this two-period setting without initial debt equals the public debt level. In the following we refer to d as the *deficit*. While the government observes the state of the economy, the legislature does not observe it directly but receives a signal x about the underlying state. The signal can take two values with the following probabilities:

$x = x_L$ with probability 1 if $y_1 = y_L$ and probability q if $y_1 = y_H$,
$x = x_H$ with probability 0 if $y_1 = y_L$ and probability $1 - q$ if $y_1 = y_H$,

where $x_L < x_H$.[8] Hence, a bad state of the economy always results in a low signal for x. However, with some probability a good state of the economy may also lead to a low signal. The parameter q reflects a Type II error (i.e., a false negative) in probabilistic terms.

The timing of events is as follows. First, the period-1 state of the economy materialises and is observed by the government. Then, the government chooses a level of fiscal deficit. Third, the legislature receives the signal x.

[7] To keep the focus on the role of a fiscal rule and transparency through the introduction of a (sufficiently strong) IFC, which we introduce below, we assume that the re-election probability is exogenous. Relaxing this assumption would complicate the analysis substantially as the incumbent government would use the deficit to strategically manipulate the voters by using the deficit as a signal. Examples of analyses that endogenize re-election chances are Poplawski-Ribeiro and Beetsma (2008) and Sloof et al. (2026).

[8] For illustrative purposes, we have chosen to rank the signals, though for formal modeling purposes this is not needed.

Fourth, the election takes place. Finally, debt is paid off, and the government incurs a potential loss from violating the rule.

We start by introducing some elements that are useful for the ensuing analysis. The socially-optimal values of the deficit in the two possible states are obtained by maximizing society's utility function under perfect information, hence they follow from the first-order condition:

$$u'(y_1 + d) = 1. \qquad (5)$$

The solutions are denoted by d_L^* and d_H^* when the economy is in a bad and a good state, respectively. We assume that $d_H^* < \bar{d} < d_L^*$. The legislature is willing to let the deficit unpunished go above $\bar{d}$ up to the level d_L^* if it deems it plausible that the economy is in a bad state. The margin between $\bar{d}$ and d_L^* is analogous to the requirement in the EU that, if the deficit exceeds the 3 percent reference level, the deviation should be small for an Excessive Deficit Procedure not to be started. However, a deficit above d_L^* will always lead to punishment for transgressing the rule.

The government's optimal deficit choice in the absence of a fiscal rule follows from the first-order condition:

$$u'(y_1 + d) = \pi. \qquad (6)$$

We denote the solutions d_L^g and d_H^g when the economy is in a bad and a good state, respectively, where we assume that $d_H^g \geq d_L^*$. This requires the probability of re-election π to be sufficiently low.

Because the legislature does not observe the state of the economy directly, it will base its assessment on the signal about its underlying actual state. Hence, the government incurs an expected cost πc associated with the deficit d exceeding $\bar{d}$ when the signal is $x = x_H$, but not if it sets $\bar{d} < d \leq d_L^*$ and the signal is $x = x_L$. Hence, in the presence of the deficit rule, the government makes the following choices. If $y_1 = y_L$, the government chooses between the socially optimal deficit of the lower state ($d = d_L^*$), implying expected government utility $u(y_L + d_L^*) + \pi(\bar{y} - d_L^*)$, and a higher deficit ($d = d_L^g$), implying expected government utility $u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)$.[9] The choice of a larger deficit beyond the low-state threshold, i.e., $d > d_L^*$, will inevitably lead the government to incur a cost of violating the fiscal rule on the deficit ceiling. If $y_1 = y_H$, the government chooses between $d = \bar{d}$, implying expected government utility $u(y_H + \bar{d}) + \pi(\bar{y} - \bar{d})$; and $d = d_L^*$, leading to expected government utility $u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1 - q)c)$, where $(1 - q)$ is the probability that the signal about the economy is positive; and versus $d = d_H^g$, implying government utility $u(y_H + d_H^g) + \pi(\bar{y} - d_H^g - c)$.

A fiscal council analyses the state of the economy. Hence, establishing a fiscal council essentially means to make the information signal x_1 more precise, i.e., it implies a reduction in the probability of a Type II error (false negative) on the probabilistic term q. A stronger fiscal council can be interpreted as a larger reduction in q, for example because it has more resources to better analyse the state of the economy or the ability to distil the true state from the signal. We analyse two cases: (i) Case A in which the government prefers not to have a fiscal

[9] Note that also d_L^g maximises the objective function $u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)$, as πc is treated as a constant.

council, but the legislature does; and (ii) Case B in which neither the government nor the legislature would prefer to have a fiscal council.[10]

Case A: $(\bar{d}, d_L^g, c)$ is such that d_L^g is strictly dominated by $\bar{d}$ in government utility.

This assumption holds if the concavity in $u(.)$ is sufficiently strong (hence, in the case of constant relative risk aversion utility, relative risk aversion is sufficiently strong) and c is sufficiently high.

For the case of a bad state of the economy, define $\bar{c}$ as the cost such that:

$$u(y_L + d_L^*) + \pi(\bar{y} - d_L^*) = u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c) \quad (7)$$

Hence, for $c > \bar{c}$, $LHS(1) > RHS(1)$, and vice versa for $c < \bar{c}$, where LHS is shorthand for "left-hand side" and RHS is shorthand for "right-hand side". For the case of a good state of the economy, define $\bar{q}(c)$ as the value of q such that:

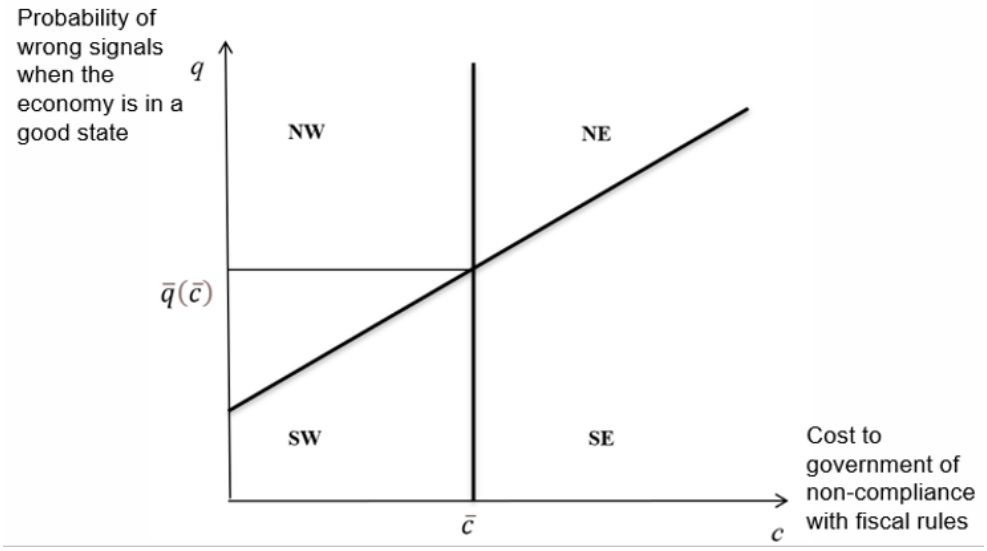
$$u(y_H + \bar{d}) + \pi(\bar{y} - \bar{d}) = u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c) \quad (8)$$

We assume that $u(.)$ is such that there exists a combination $(\bar{c}, \bar{q}(\bar{c}))$ that makes both (7) and (8) hold. Because $u(y_H + d_L^*) > u(y_H + \bar{d})$, this is the case, for example, if π and c are sufficiently low. The function $\bar{q}(c)$ is increasing in c, implying a stronger rule allows for a lower strength of the council (less precision) to induce a shift from deficit d_L^* to deficit $\bar{d}$ in a good state.

Figure 3 defines four areas: (i) area North-West (NW), where $LHS(7) < RHS(7)$ and $LHS(8) < RHS(8)$; (ii) area South-West (SW), where $LHS(7) < RHS(7)$ and $LHS(8) > RHS(8)$; (iii) area North-East (NE), where $LHS(7) > RHS(7)$ and $LHS(8) < RHS(8)$; and (iv) area South-East (SE), where $LHS(7) > RHS(7)$ and $LHS(8) > RHS(8)$.

[10] To keep matters tractable, in the following, we ignore the possibility that the legislature may update the probability it attaches to the different states of the world on the basis of the observed deficit and, based on this updating possibility, set the parameter combination $(\bar{d}, c, q)$ characterizing the institutional arrangement to maximise its utility. The government would internalize the implications in terms of the expected punishment when setting the deficit. The equilibrium concept would be that of a Perfect Bayesian Equilibrium, which could in general give rise to pooling equilibria in which both the "high-type" and "low-type" government (in terms of period 1 income) choose the same deficit, separating equilibria in which both types select different deficit levels, and mixed-strategy equilibria. Such an exploration would go beyond the current paper, which introduces the theoretical framework primarily to help interpret the findings of the ensuing empirical analysis, and is left for future research. A related theoretical analysis in a model analysing optimal (strict) debt ceilings is found in Sloof et al. (2026).

Figure 3. Graphical Illustration of the Model



Source: Authors' illustration.
Note: The figure illustrates four different quadrants that reflect the interaction of fiscal councils and rules according to the extent of asymmetric information (probability q about the signals on the economy) and the cost of government in noncomplying with fiscal rules (c). NW = Northwest quadrant; NE = Northeast quadrant; SW = Southwest quadrant; SE = Southeast quadrant.

Start in NW. Hence, the government's expected and legislature's expected utilities are:

$$\lambda\,[u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)] + (1-\lambda)[u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c)] \quad (9)$$

and

$$\lambda\,[u(y_L + d_L^g) + (\bar{y} - d_L^g)] + (1-\lambda)[u(y_H + d_L^*) + (\bar{y} - d_L^*)] \quad (10)$$

The introduction of a (sufficiently strong) fiscal council amounts to a reduction in q. Such a reduction lowers government's expected utility up to the point of transition into SW, in which case it becomes:

$$\lambda\,[u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)] + (1-\lambda)\left[u\left(y_H + \bar{d}\right) + \pi\left(\bar{y} - \bar{d}\right)\right] \quad (11)$$

The government's expected utility remains constant at the switch point because there $LHS(2) = RHS(2)$. Any further reduction in q has no effect on the government's expected utility. The legislature's expected utility upon entry into SW is:

$$\lambda\,[u(y_L + d_L^g) + (\bar{y} - d_L^g)] + (1-\lambda)\left[u\left(y_H + \bar{d}\right) + \left(\bar{y} - \bar{d}\right)\right] \quad (12)$$

Because $d_H^* < \bar{d} < d_L^*$, the legislature's expected utility is maximised at $d = d_H^*$ for $y_1 = y_H$, and $u(.)$ is strictly increasing and strictly concave, the legislature's expected utility jumps to a higher level when transitioning from NW to SW. In other words, starting in NW the introduction of a sufficiently strong fiscal council or a sufficient

strengthening of an existing fiscal council that brings the economy into SW lowers the government's expected utility, but it raises the legislature's expected utility. This creates a conflict between the government and the legislature. If the government prevails in this conflict, then no fiscal council or a weak fiscal council will be installed, while if the legislature prevails, a sufficiently strong fiscal council will be installed or a given fiscal council will be strengthened to a sufficient extent.

Now, start in area NE. In this case, because $LHS(7) > RHS(7)$ and $LHS(8) < RHS(8)$, expected government utility is:

$$\lambda\,[u(y_L + d_L^*) + \pi(\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c)] \qquad (13)$$

while the legislature's expected utility is:

$$\lambda\,[u(y_L + d_L^*) + (\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^*) + (\bar{y} - d_L^*)] \qquad (14)$$

A reduction in q such that the system stays in NE lowers the government's expected utility but keeps the legislature's expected utility unchanged. If the reduction in q is so large that the system transitions from NE to SE, the government's and legislature's expected utilities switch to, respectively:

$$\lambda\,[u(y_L + d_L^*) + \pi(\bar{y} - d_L^*)] + (1-\lambda)\big[u\big(y_H + \bar{d}\big) + \pi\big(\bar{y} - \bar{d}\big)\big] \qquad (15)$$

and

$$\lambda\,[u(y_L + d_L^*) + (\bar{y} - d_L^*)] + (1-\lambda)\big[u\big(y_H + \bar{d}\big) + \big(\bar{y} - \bar{d}\big)\big] \qquad (16)$$

Any further reduction in q has no effect on the government's or legislature's expected utility. Because $d_L^* > \bar{d} > d_H^*$, the legislature's expected utility is maximised at $d = d_H^*$ for $y_1 = y_H$, and $u(.)$ is strictly increasing and strictly concave, the legislature's expected utility jumps to a higher level when transitioning from NE to SE. The reason is that the deficit in the good state falls from d_L^* to $\bar{d}$. In other words, starting in NE the introduction of a sufficiently strong fiscal council or a sufficient strengthening of an existing fiscal council lowers the government's expected utility, but it raises the legislature's expected utility. Again, there is a conflict between the government and the legislature. If the government prevails, then no fiscal council or a weak fiscal council will be installed, while if the legislature prevails, a sufficiently strong fiscal council will be installed or a given fiscal council will be strengthened to a sufficient extent.

Notice that, starting in NW or NE, we observe that a sufficiently strong reduction in q lowers the deficit outcome from d_L^* to $\bar{d}$ in the good state. In other words, the introduction of a sufficiently strong fiscal council increases the likelihood that the fiscal rule will be adhered to.

Further, notice that when in SW or SE a marginal tightening of the deficit ceiling $\bar{d}$ predicts a fall in the deficit when the state of the economy is good, as the deficit itself in this case is $\bar{d}$. The deficit in the case of a bad state of the economy is unchanged. Hence, when in SW or SE a marginal tightening of the deficit ceiling increases the legislature's expected utility by pushing the deficit itself in the direction of d_H^*, its optimum when the state is good.

Case B: $\big(\bar{d}, d_L^g, c\big)$ is such that $\bar{d}$ is strictly dominated by d_L^g in government utility.

This assumption holds if the concavity in $u(.)$ is sufficiently mild and c is sufficiently low. Again, we can divide the (c, q) space into four areas and analyse this case in the same way as Case A. Annex B contains the detailed analysis. With $\bar{d}$ strictly dominated by d_L^g, the comparison in the good state switches from (8) to (B.2) in Annex B. Starting and staying in NW (NE), a reduction in q lowers the government's expected utility but keeps the legislature's expected utility constant. However, at the transition from NW to SW (NE to SE) the legislature's expected utility jumps to a lower level, as in the good state of the economy the deficit rises even further (from d_L^* to d_L^g), while the government's expected utility stays unchanged. As an aside, in Case B the government never sets the deficit at $\bar{d}$.

Cases A and B lead to different outcomes when it comes to choosing whether to install a fiscal council or not. The relevance of Case A for some countries and that of Case B for other countries is in line with the second stylized fact listed at the beginning of this section. They also lead to different conclusions about the usefulness of installing a deficit rule with a reference value of $\bar{d}$. In Case B the rule fulfils effectively no role, because it has no effect on fiscal outcomes. This case could potentially account for why some countries do not feature a fiscal rule, in line with the first stylized fact listed at the beginning of this section. For these countries, installing a rule may have no effect on fiscal policy in any case, possibly because for political reasons the cost c of violating a rule can be set at most at a very low level, making the presence of a rule a mere nuisance. Further, in both Cases A and B, there is only a role for a fiscal council when fiscal rules are present, as a council helps to reduce the likelihood of a wrong signal when the state is good, which seduces the government into violating the rule. This is consistent with the third stylized fact above that councils tend to be established once a rule is already in place. Our ensuing empirical analysis, however, focuses largely on the consequences of making existing fiscal rules and councils stronger, suggesting that Case A is the more relevant case for our empirical analysis. Therefore, in the following we assume that Case A applies.

Overall, the framework generates testable empirical hypotheses on the effects of strengthening a fiscal rule, introducing or strengthening a fiscal council, and simultaneously strengthening a fiscal rule and introducing or strengthening a fiscal council. First, consider the effect of a stronger deficit rule. Starting in the region NW with no or a relatively weak IFC, a sufficiently large increase in the cost c brings the economy into region NE. The transition implies in a bad state a shift from d_L^g to d_L^*, while the good-state deficit outcome is unchanged. A further increase in cost c brings the economy from region NE into the region SE. Now the deficit in a bad state is unchanged, while it falls in a good state. Hence, the expected deficit falls when transitioning through the regions when the cost c rises enough. Starting in region NW with a not too weak IFC, i.e. $q < \bar{q}(\bar{c})$ (but q not too low), increasing cost c brings the economy first into the region SW, leading to an unchanged deficit when the state is bad, but a fall in the deficit when the state is good. Increasing the cost further brings the economy into region SE, where the deficit is unchanged in the good state, but falls in the bad state. Hence, the expected deficit falls when transitioning through the regions when the cost c rises enough. Finally, starting in region SW, increasing cost c brings the economy into the region SE, again leading to a fall in the expected deficit. Overall, this leads us to the following proposition:

Hypothesis 1: Starting without, or with a sufficiently weak, deficit rule and making the rule sufficiently stronger (raising c) would reduce the expected deficit.

We now turn to the effect of introducing a fiscal council or strengthening it to a sufficient extent:

Hypothesis 2: Starting without, or with a sufficiently weak, council and for a given strength of the deficit rule c, including zero (no deficit rule), the introduction, or strengthening, of a fiscal council would reduce the expected deficit.

Hypothesis 3: In the absence of a deficit rule or in the presence of a sufficiently weak rule and no fiscal council or a sufficiently weak council,[11] the introduction (or sufficient strengthening) of a rule combined with the introduction (or sufficient strengthening) of a council reduces the expected deficit more than by merely either introducing (or sufficient strengthening) of a rule or introducing (or sufficient strengthening) of a fiscal council.

In other words, Hypothesis 3 makes a prediction about the *interaction* between the strength of a deficit rule and the presence or strength of a fiscal council for the deficit itself. Both instruments complement each other in lowering the expected deficit. The next section will empirically test these hypotheses. Moreover, we will also check how (strong) fiscal rules and councils impact primary deficits' forecasting errors and the accuracy of their forecasts, which will be the focus of the subsequent section.

We now turn to the forecasting errors and their accuracy. Because expectations are rational, the framework predicts that the forecasting errors are on average zero in each of the quadrants of the (c, q) space. Hence, a strengthening of the rules or of the fiscal council will not affect the average forecasting error. This is not the case for the accuracy of the forecasts which depends on the institutional setting (c, q). However, the predicted impact of a stronger rule or council on the forecasting accuracy is ambiguous and depends on the specific starting location in the (c, q) space. Some algebra shows that for Case A the variance of the prediction error is:

$$(1-\lambda)\,(1-q)[\lambda+(1-\lambda)q](d_{BS}-d_{GS})^2 \qquad (17)$$

where d_{BS} and d_{GS} are the deficit outcomes in the bad, respectively good, state. In particular, $(d_{BS}, d_{GS}) = (d_L^g, d_L^*)$ in NW, $(d_{BS}, d_{GS}) = (d_L^g, \bar{d})$ in SW, $(d_{BS}, d_{GS}) = (d_L^*, d_L^*)$ in NE and $(d_{BS}, d_{GS}) = (d_L^*, \bar{d})$ in SE. Starting in NW and making the rule stronger—i.e., increasing the cost c—the variance of the forecast error falls when the economy transitions into NE, but it jumps again to a higher level when the economy transitions from NE to SE. Starting in SW and strengthening the rule so as to transition into SE, leads to a fall in the forecast error variance. To show that the effect of a marginally stronger council, i.e. a marginal reduction in q, is ambiguous, we can take the first- and second-order derivatives of the common term $(1-\lambda)\,(1-q)[\lambda+(1-\lambda)q]$ with respect to q, to see that for $\lambda < 1/2$ this term is first increasing and then decreasing in q.[12] Larger reductions in q that cause the economy to transition from NW to SW cause the variance of the projection error to fall, while a transition from NE to SE causes it to rise.

[11] That is, we start in NW and end in SE.

[12] As a special case consider the extreme case in which q is 1 and the economy is area NW in Case A. The government will always set the deficit to d_L^g and anticipating this, this is also what is projected. Accuracy is at its maximum, despite the council being completely uninformative in its assessment.

IV. The Relationship between Primary Deficits and Fiscal Rules and Councils

This section explores the role of fiscal rules and councils in driving primary deficit outcomes. Violating fiscal rules may be costly for a government, for example, because it makes a government look less competent in the eyes of the general public or the financial markets, which is related to the reputational costs discussed in our theoretical framework above, with potential political consequences for the government. In some cases, as in the case of the EU, financial sanctions may also be imposed if the reference values for the public deficit or the public debt are violated and the government fails to correct the transgression. In the above theoretical framework, these costs of violating the rules were captured by the cost c. Fiscal councils also make it more costly for governments to behave in a fiscally undisciplined way. For example, through their analyses and the publication of these analyses, they may highlight the role of the government itself relative to exogenous factors beyond its control when assessing fiscal outcomes. Also, fiscal councils may be heard in Parliament about fiscal policy.

IV.1. The regression framework

To analyze the effect of rules and councils on fiscal outcomes, we use the following general format for the regression framework:

$$d_{it} = C_i + D_t + \sum_{\tau=1}^{T} \rho_\tau d_{i,t-\tau} + \xi DS_{it} + \sum_{\tau=0}^{T} \alpha_\tau FR_{i,t-\tau} + \sum_{\tau=0}^{T} \beta_\tau FC_{i,t-\tau} + \sum_{\tau=0}^{T} \gamma_\tau \left(FR_{i,t-\tau} * FC_{i,t-\tau}\right) + \sum_{\tau=0}^{T} \delta_\tau CONTROLS_{i,t-\tau} + \varepsilon_{it}, \quad (18)$$

where d_{it} is the primary deficit of country i in period t; C_i is a country-fixed effect; D_t is a time-fixed effect; $FR_{i,t-\tau}$ is a fiscal rule variable (e.g., a dummy for the presence of a fiscal rule, a dummy for the presence of a fiscal rule of a certain type, the strength of the rule, etc.); $FC_{i,t-\tau}$ is a fiscal council variable (e.g., a dummy for its presence, the strength of the council, etc.); DS_{it} is a dummy that takes a value of one if one or more of the fiscal rules are suspended in country i, and zero if not; and $CONTROLS_{i,t-\tau}$ is a set of controls capturing macroeconomic variables, uncertainty, etc.

While Equation (18) includes indicators for fiscal rules and fiscal councils, throughout the paper, we use different combinations of these indicators, starting by estimating the association with rules or councils separately and then combining and/or interacting them. For $FR_{i,t-\tau}$ capturing fiscal rule strength, *Hypothesis 1* predicts that $\sum_{\tau=0}^{T} \alpha_\tau < 0$. For $FR_{i,t-\tau}$ capturing a fiscal rule dummy and $FC_{i,t-\tau}$ capturing fiscal council strength, *Hypothesis 2* predicts that $\sum_{\tau=0}^{T} \beta_\tau < 0$. Finally, for $FR_{i,t-\tau}$ and $FC_{i,t-\tau}$ capturing fiscal rule strength, respectively fiscal council strength, *Hypothesis 3* predicts that $\sum_{\tau=0}^{T} \gamma_\tau < 0$. The fiscal and macroeconomic variables used in these regressions are taken from the April 2025 vintage of the WEO. In all regressions we include year- and country-

fixed effects (not separately reported), the lagged primary deficit,[13] the lagged change in the debt ratio over GDP, real GDP growth, and consumer price inflation. Real growth helps to capture the role of the business cycle.

IV.2. The instrumentation strategy

Our explanatory variables may be endogenous to the primary deficit because we would expect an increase in the primary deficit to stimulate growth and inflation. Also, the adoption of fiscal rules and fiscal councils may be a response to the fiscal performance of a country. On the one hand, the adoption of fiscal rules and fiscal councils could be a way to strengthen discipline where this is difficult to achieve otherwise. On the other hand, a strong culture of fiscal discipline may make it politically easier to adopt (strong) rules and councils. Hence, we estimate all our models using the Two-Stage Least Squares estimator for panel data with fixed effects. The instrument set comprises weighted averages of growth, inflation, and the included rule and/or council variables of all other countries in the system. More specifically, it is a distance-weighted (in logs) spatial lag of other countries' corresponding variables.[14,15] Here, these weighted averages are proportional to the log-distance between the capitals of the countries included in the sample. That is, the instrument for variable x is given by $instr_x_{it} = \sum_{j \neq i} \left(\frac{log(dist_{ij}) x_{jt}}{\sum_{j \neq i} log(dist_{ij})} \right)$,

where $dist_{ij}$ is the distance between the capitals of countries i and j. The rationale behind the construction of this instrument is that we expect relatively little feedback from an individual country's fiscal performance onto these weighted averages across a large set of other countries. Any reverse causality would moreover be diluted further by giving a larger weight to countries further away from the country of which the fiscal performance is to be explained.

We assess the adequacy of these instruments using two complementary diagnostics. The Kleibergen-Paap (2006) rk Wald F-statistics, a first-stage weak-identification diagnostic, generally exceeds 40 in specifications without interactions and is between 20 and 30 for those with interactions.[16] These values provide descriptive evidence against weak identification, although no generally applicable critical values exist for the cluster-robust Kleibergen and Paap statistics. Second, we estimate Equation (18) with OLS regression while adding the instruments to the set of explanatory variables (Online Annex B). In most instances the instrument and the corresponding regressor are not statistically significant simultaneously, suggesting that the instruments only affect the dependent variables through the regressors.

[13] Given that this is a dynamic regression, the coefficients on the lagged primary deficits (ρ_τ) can be used to compute the long-run effect of the fiscal rule or council by scaling its short-run coefficient by $1 / (1 - \sum_{\tau=1}^{T} \rho_\tau)$, yielding a larger total impact if $\sum_{\tau=1}^{T} \rho_\tau > 0$ than their short-run coefficients (α_τ and β_τ) alone.

[14] The construction of our instruments is related to, for example, the spatial-econometrics instruments deployed by Foucault et al. (2008) and the contiguity instrument of Caselli and Reynaud (2020).

[15] As just discussed, the endogeneity of fiscal rules and councils to a country's fiscal performance is challenging to address. So, even with the instrumentation strategy here proposed, endogeneity remains a potential threat and would call for more in-depth analysis beyond the scope of this paper. For example, potentially stronger instruments could use political and institutional variables that could identify more accurately the relationship between fiscal performance and rules and councils. The distinction between rules and councils that have only been recently implemented or exist for some time, hence have become a given in a country's fiscal policy, is another possibility to be explored.

[16] First-stage regressions are reported in Online Annex A.

IV.3. Baseline results

Table 2 reports panel estimates of Equation (18) using only indicators for fiscal rules (and not fiscal councils) for our country sample. This table focuses on the role of fiscal rules and their strength. The first lag of the primary deficit always enters highly significantly with a coefficient of 0.57 or 0.59. This reflects the fact that fiscal variables generally exhibit substantial persistence. The lagged change in the public debt ratio generally has no effect. Further, a one-percentage point increase in growth is associated with a highly significant fall in the primary deficit, by 0.15 percentage point of GDP. It is consistent with the fact that higher growth, by generating more government revenues, is often associated with a lower primary deficit.[17] The estimates further indicate that a one percentage point higher inflation rate is associated with a lower primary deficit, by 0.03 percentage point of GDP. Although economically small, the effect is statistically highly significant. Potential explanations are the presence of certain forms of financial repression that make it hard to escape the effects of higher inflation. In addition, the lack or presence of indexation to inflation in fiscal revenue and expenditure items may affect the deficit (Balasundharam et al., 2023). For example, there may be a role for "bracket creep"—i.e., tax brackets are not fully indexed to inflation. We also control for the fact that during crises numerical fiscal rules may be temporarily suspended, such as by invoking an escape clause or by parliamentary approval. For example, during the COVID-19 pandemic the EU fiscal rules were suspended while at the same time also national escape clauses were activated in some countries.

Columns (2) and (3), in which fiscal rule presence and rule strength enter the regression separately, provide an initial indication that the strength of a fiscal rule matters more than its mere presence. The fiscal rule dummy is statistically insignificant in Column (2), whereas the coefficient on rule strength is negative and highly significant in Column (3). The suspension dummy is positive and highly significant in both specifications.

In Column (4) we enter the presence of at least one fiscal rule (captured with a dummy) and the strength of the fiscal rules in place. While rule presence is not significant, rule strength is estimated to have a highly significant negative effect on the primary deficit.[18] This is in line with *Hypothesis 1* based on our theoretical framework where an increase in cost c—i.e., a strengthening of the rule, produces a lower expected deficit.[19] A qualitatively similar pattern is observed for the combination of a dummy for the presence of a debt rule and its strength (Column (5)) and the combination of a dummy for the presence of a budget balance rule and its strength (Column (6)). To give an indication of the economic importance of the estimates, we consider the effect of shifting from the 25th to the 75th percentile of the distribution of fiscal rule strength. Here, and in the sequel, the distribution is based on all observations in which a rule (or, later, a council) is present. This amounts to a shift of 0.35 on a 0–1 scale, suggesting it is associated with a reduction of the primary deficit by 1.2 percentage points of GDP, based on the estimate in Column (4). To put this into a broader perspective: this amounts to almost 60 percent of the gap

[17] The coefficients on institutional variables remain broadly unchanged when GDP growth is replaced with the output gap (Online Annex Table C.6). Unlike growth, the output gap is not statistically significant in those regressions.

[18] We also estimated Equation (18) including only the dummy for fiscal rules' presence and without rule strength. Online Annex Table C.1 shows that the dummy for rules' presence remains insignificant, but both a dummy for the presence of a debt rule or a budget balance rule are significant and with the correct (negative) sign.

[19] These results could potentially be confounded by concurrent improvements in public financial management (PFM). However, such improvements may themselves be a consequence of fiscal rule adoption, as suggested by Hatchondo et al. (2022a, 2022b). Moreover, recent empirical evidence points to only moderate PFM improvements over the past two decades (Rahman et al., 2026).

between the 25th percentile of the primary deficit (–0.14 percent of GDP) and the 75th percentile (1.98 percent of GDP) for the full sample of countries. Using the coefficient on the lagged primary deficit in that column and accounting for the persistence of primary deficits (see Footnote 13), the implied long-run effect is larger—about 2.3 times the short-run impact—corresponding to a reduction in the primary deficit of about 2.8 percentage points of GDP. Finally, in periods in which fiscal rules are suspended, primary deficits are about 1.2 percentage points of GDP higher on average than when fiscal rules are active. The effect is statistically highly significant. It is not necessarily exclusively the direct result of the inactivity of the rules, however, since it could also be the result of adverse economic and financial circumstances.[20]

Table 2. Regressions of the Primary Deficit on Fiscal Rules, 2000–2024

Dependent variable: Primary Deficit	(1)	(2)	(3)	(4)	(5)	(6)
Lag Dependent Variable	0.59***	0.59***	0.57***	0.57***	0.57***	0.57***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Lag Δ Debt	0.02*	0.02*	0.02	0.02	0.02	0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Real Growth	-0.15***	-0.15***	-0.15***	-0.15***	-0.15***	-0.15***
	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Inflation	-0.03***	-0.03***	-0.03***	-0.03***	-0.03***	-0.03***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Fiscal Rule Suspension Dummy		0.79***	1.19***	1.19***	1.15***	1.18***
		(0.17)	(0.18)	(0.18)	(0.19)	(0.19)
Fiscal Rule Dummy		-0.36		-0.02		
		(0.27)		(0.28)		
Fiscal Rule Strength			-3.44***	-3.43***		
			(0.62)	(0.67)		
Debt Rule Dummy					0.07	
					(0.36)	
Debt Rule Strength					-2.86***	
					(0.59)	
Budget Balance Rule Dummy						0.50*
						(0.30)
Budget Balance Rule Strength						-3.23***
						(0.67)
R-squared	0.53	0.53	0.53	0.53	0.53	0.53
F test statistics	75.07	86.05	81.03	78.36	77.05	78.36
F test p-value	0.00	0.00	0.00	0.00	0.00	0.00
KP LM test p-value	0.00	0.00	0.00	0.00	0.00	0.00
KP Wald F test statistics	78.28	39.46	45.07	45.49	52.21	45.49
Observations	2,058	2,058	2,058	2,058	2,058	2,058
Number of countries	90	90	90	90	90	90
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes

Sources: Authors' estimations. See Annex A for all data sources used in the estimations.
Notes: Kleibergen-Paap (2006) rk LM, Hansen J, and Kleibergen-Paap (2006) rk Wald F-statistics included. The binary variable 'Fiscal Rule Suspension Dummy' takes the value of 1 if at least one fiscal rule is suspended. Standard errors are clustered at the country level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

[20] An alternative formulation would interact the dummy with the rule's variables. However, we have chosen for the current formulation as it is conceivable that, because of inertia in fiscal decision-making rules still exert some influence even during periods when they are suspended.

IV.4. Entering fiscal councils into the baseline

Table 3 enters the presence of a fiscal council, its strength, and fiscal rules strength, individually and through interaction terms between the latter and the fiscal council variables, in the primary deficit regression. Unlike the case of rules in Table 2, we do not include a regression in which we include council presence and strength jointly, as council strength does not vary over time. The coefficient estimates of the lagged primary deficit, real growth and inflation remain virtually unchanged. The coefficient estimate of the fiscal suspension dummy remains almost the same as before and highly statistically significant. Column (1) shows that fiscal rule strength enters, as expected, with a highly significant negative sign, again in line with *Hypothesis 1*. Replacing rule strength with a council dummy (Column (2)) or council strength (Column (3)), the two variables enter with a highly statistically significant and negative sign, in line with *Hypothesis 2*.[21] A shift from the 25th to the 75th strength percentile for fiscal councils—i.e., an increase by 0.23 on a 0–1 scale, would be associated with a reduction of the primary deficit by 0.42 percentage point of GDP. When combining the variables on fiscal rule strength, the council dummy, and the interaction of both (see Column (4)), fiscal rule strength retains its original magnitude and remains highly significant. However, the council dummy is not significant, nor is its interaction term with the fiscal rule strength. Replacing the council dummy with council strength in Column (5) yields the same qualitative pattern. Our finding that the interaction effects are not significant may be read as evidence against *Hypothesis 3* above. There are some plausible explanations for this result. One is that the hypothesized interaction effects are not really there. In particular, in practice rule strength and council strength may each be doing the work already of the other, that is, they substitute for each other in their effect on fiscal discipline. A second explanation is that the fiscal councils in our dataset are so different from each other, and therefore in the way they can leverage the effect of rule strength, it is hard to trace an effect of the interaction between rule and council strength in general. Third, in view of the limited variation in fiscal council data over the sample, by including all three variables together, we may be asking “too much” from the data. We may simply have too few data, especially on fiscal councils, to produce statistical significance of the interactions.

[21] As a robustness check, we also add a control variable for years since the establishment of a fiscal council to capture potential differences between early and more mature stages. The results (not shown here and available upon request) indicate that the associated coefficient is not statistically significant.

Table 3. Regressions of the Primary Deficit on Rules, Councils and their Interaction, 2000–2024

Dependent Variable: Primary Deficit	(1)	(2)	(3)	(4)	(5)
Lag Dependent Variable	0.57***	0.58***	0.58***	0.57***	0.57***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Lag Δ Debt	0.02	0.02*	0.02*	0.02	0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Real Growth	-0.15***	-0.15***	-0.15***	-0.15***	-0.15***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Inflation	-0.03***	-0.03***	-0.03***	-0.03***	-0.03***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Fiscal Rule Suspension Dummy	1.19***	1.01***	1.01***	1.19***	1.19***
	(0.18)	(0.19)	(0.18)	(0.18)	(0.18)
Fiscal Rule Strength	-3.44***			-3.93***	-3.70***
	(0.66)			(1.05)	(0.99)
Fiscal Council Dummy		-1.26***		0.19	
		(0.33)		(0.55)	
Fiscal Council Strength			-1.79***		0.50
			(0.48)		(0.92)
Fiscal Rule Strength x Fiscal Council Dummy				0.18	
				(1.28)	
Fiscal Rule Strength x Fiscal Council Strength					-0.27
					(1.82)
R-squared	0.53	0.53	0.53	0.53	0.53
F test statistics	81.03	75.50	76.36	78.53	78.66
F test p-value	0.00	0.00	0.00	0.00	0.00
KP LM test p-value	0.00	0.00	0.00	0.00	0.00
KP Wald F test statistics	67.82	63.38	42.86	25.82	30.08
Observations	2,058	2,058	2,058	2,058	2,058
Number of countries	90	90	90	90	90
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Country fixed effect	Yes	Yes	Yes	Yes	Yes

Sources: Authors' estimations. See Annex A for all data sources used in the estimations.
Note: Kleibergen-Paap (2006) rk LM, Hansen J, and Kleibergen-Paap (2006) rk Wald F-statistics included. The binary variable 'Fiscal Rule Suspension Dummy' takes the value of 1 if at least one fiscal rule is suspended. Standard errors are clustered at the country level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Hansen J statistics are not reported when the number of clusters is insufficient to calculate a robust variance-covariance matrix.

IV.5. Robustness Checks and Additional Tests

In this section we explore the robustness of the results by expanding the baseline with a number of extra controls. We do not include these controls in the baseline above, because the data are generally not complete, implying that our sample shrinks.

The first additional control tested is the change in the 10-year government bond spread.[22] Besides exploring whether the association of the primary deficit with our institutional variables is unchanged, it is interesting to

[22] The government bond spread is calculated as the difference between the 10-year government bond yield of each country and the corresponding 10-year U.S. government bond yield.

check whether governments engage in budgetary contraction when yields and, hence, future interest payments on the public debt go up. However, this is not borne out by the estimates presented Online Annex Table C.2. The change in the 10-year yield exerts a positive effect that is significant at the 10 percent level in some specifications. The results on our institutional variables become slightly stronger, with both rule and council strength exerting a negative effect on the primary deficit, in line with *Hypotheses 1* and *2*, although council strength is not significant. We suspect that this is not so much the effect of controlling for yield changes, but rather the fact that the full sample now excludes countries with weaker data availability, while the sample is still of large size.

The second robustness test adds political variables as additional controls in the estimations. Online Annex Table C.3 reports a dummy equaling 1 in election years that is never statistically significant. Yet, the coefficients on our institutional variables of interest remain essentially unaffected. A second set of estimations (Online Annex Table C.4) adds instead the number of changes in government in a given year. The coefficients for that political variable indicate a positive effect that is significant at the 5-percent level. These results suggest that the effect of one change in government in a year leads to a higher primary deficit of around 0.20 percentage point of GDP. Less political stability as exhibited by more frequent government turnover likely implies less fiscal control. Importantly, the association with the fiscal institutions remains by and large unchanged.

A third test investigates whether economic uncertainty impacts the estimations, in particular of the effect of our fiscal institutional variables (see Online Annex Table C.5). Using primary deficit as dependent variable, economic uncertainty is not statistically significant in any of the estimations. Hence, we do not include it in the baseline regressions. The association with our rule and council variables remains unchanged.

As a fourth test, we replaced country fixed effects with fixed effects by geographical region, maintaining time fixed effects. Qualitatively, the association with the strengthening of a rule remains the same, but the quantitative effects become substantially weaker. Replacing year fixed effects with region-year fixed effects, while maintaining country fixed effects, we find that the effect of debt rule strength and budget balance rule strength disappears. Dropping the EU from the sample also generally weakens the association with rule strength.

We close this section by digging deeper into the sources of the effects of fiscal institutions on the primary deficit. We do this by splitting the latter into primary spending and revenues. The results are reported in Online Annexes D and E, respectively. Interestingly, when included individually our fiscal institutional variables are only statistically significant for revenues (see Online Annex Table D.1) and not for primary expenditure (Online Annex Table E.1). These results suggest that the negative association of the primary deficit with stronger rules primarily originates from the fact that stronger rules and stronger councils are associated with an increase in revenues (and not a reduction in primary expenditure). For the case in which rule strength, council variables and their interactions are included, the coefficients on the latter two work to counteract the negative effect rule strength on primary expenditure.

V. The Relationship between Forecasting Errors and Accuracy with Rules and Councils

This section expands the empirical analysis by investigating the effect of fiscal rules and councils on primary deficit forecasting errors and the accuracy of the forecasts. The theoretical framework predicts a zero-forecasting error on average, while the effect of (strengthening) a rule or council on forecasting accuracy is ambiguous, which prevents us from formulating meaningful hypotheses on these relationships. Nevertheless, we are interested in the empirical performance of rules and councils for forecasting errors and accuracy. Fiscal councils generally have a role in macroeconomic forecasting and in assessing the government's macroeconomic forecasts, which makes it interesting to explore their effectiveness in this regard.

While the theoretical framework predicts average forecasting errors of zero, the literature reports widespread overoptimism bias in fiscal forecasts (e.g., Beetsma et al., 2009; Beetsma et al., 2023; and Chocobar et al., 2025), likely driven by political considerations. A government that wants to create more room for spending would have an incentive to produce overoptimistic fiscal projections (Frankel, 2011), although incentives for producing overly pessimistic forecasts are also conceivable, for example, if the minister of finance wants to rein in spending claims from line ministries. If such systematic biases indeed exist, they might be curbed by a fiscal council that improves fiscal transparency, thereby enhancing the understanding of the government's budgetary situation, inducing the government to produce more realistic projections and making it easier to hold the government accountable for systematic deviations from the budgetary projections (Merola and Perez, 2013). The effect of a (stronger) fiscal rule on such systematic projection errors is not a priori clear: governments are probably motivated to produce budgetary projections that obey the rule's numerical limit,[23] while they are also motivated to produce outcomes that obey the same limit, in order to avoid loss of reputation or punishment. The disciplining effect of fiscal rules may push fiscal outcomes into the direction of ambitious projections.

The ensuing empirical analysis examines whether fiscal councils and rules systematically affect fiscal projection errors. In the following, all forecasts for some year *t* are taken from the IMF's World Economic Outlook (WEO), which is the forecast of individual IMF country teams. Country teams often incorporate inputs from national budgets and governmental forecasts but make their own judgements regarding the macro-fiscal landscape. For example, in the context of IMF Article IV reports (country reviews), IMF staff typically acknowledge the discussions with government officials and note discrepancies between government and IMF forecasts and assessments.[24] The empirical analysis relies on the April vintage of WEO fiscal forecasts from the same year, with IMF staff usually submitting their projections around January-February. Hence, the forecast error for year *t* is constructed as the difference between the year *t* outcome taken from the April 2025 vintage of the WEO and

[23] In the case of the EU, an excessive deficit procedure may be started on the basis of the budget presented for the coming year.

[24] A key consideration for interpreting our results is the degree of independence of IMF fiscal forecasts from government forecasts. Merola and Perez (2013), analyzing 15 European countries from 1999 to 2007, find that while international agencies attempt to extract objective information from government forecasts, they are not fully immune to governments' political biases when separating true macroeconomic drivers and policy measures quantifications from politically influenced projections. The authors acknowledge that this reliance partly reflects the relatively smaller specialized staff of international organizations compared to national agencies, leading to a dependence on member states' information. Yet, other studies find IMF fiscal forecasts to be more accurate, efficient, and less rigid than private sector forecasts, though they still exhibit some optimism bias relative to realized outcomes (see An et al., 2018, for the overall balance; and Estefania-Flores et al., 2023, for the public debt).

the forecast for year *t* from the April year *t* vintage of the WEO.[25] Using WEO forecasts instead of forecasts directly from the individual countries ensures a uniformly constructed dataset that is as complete as possible.

However, forecasting also has a "technical side". Better data collection methods, more advanced models, and more capacity to produce projections may all lead to more accurate projections. The effect of (stronger) rules and councils on forecasting accuracy will be an entirely empirical question, as our theoretical framework does not provide unambiguous predictions here. The numerical limit of a fiscal rule can guide both projections and outcomes towards this limit, thereby improving forecasting accuracy. For example, using survey data, Poplawski-Ribeiro and Rülke (2011) find evidence of improved accuracy of deficit forecasting by financial experts in the case of France during the Stability and Growth Pact. Further, fiscal councils may help official macroeconomic and budgetary forecasters in more accurately assessing the state of the economy.

The regression equations in this section are essentially of the same general format as Equation (18) for the primary deficit outcome. Concretely, for the case of primary deficit forecasts, the regression becomes:

$$d_{it} - d_{i,A,t} = C_i + D_t + \sum_{\tau=1}^{T} \rho_\tau \left(d_{i,t-\tau} - d_{i,A,t-\tau}\right) + \xi DS_{it} + \sum_{\tau=0}^{T} \alpha_\tau FR_{i,t-\tau} + \sum_{\tau=0}^{T} \beta_\tau FC_{i,t-\tau}$$

$$+ \sum_{\tau=0}^{T} \gamma_\tau \left(FR_{i,t-\tau} * FC_{i,t-\tau}\right) + \sum_{\tau=0}^{T} \delta_\tau CONTROLS_{i,t-\tau} + \varepsilon_{it}, \qquad (19)$$

where $d_{i,A,t}$ is the projection in the WEO of April year *t* of the primary deficit in year *t*. $CONTROLS_{i,t-\tau}$ is a set of controls that includes among other things macroeconomic forecast errors. For the case of the accuracy of the primary deficit forecast, the regression becomes

$$Abs\left(d_{it} - d_{i,A,t}\right) = C_i + D_t + \sum_{\tau=1}^{T} \rho_\tau Abs\left(d_{i,t-\tau} - d_{i,A,t-\tau}\right) + \xi DS_{it} + \sum_{\tau=0}^{T} \alpha_\tau FR_{i,t-\tau} + \sum_{\tau=0}^{T} \beta_\tau FC_{i,t-\tau} +$$

$$\sum_{\tau=0}^{T} \gamma_\tau \left(FR_{i,t-\tau} * FC_{i,t-\tau}\right) + \sum_{\tau=0}^{T} \delta_\tau CONTROLS_{i,t-\tau} + \varepsilon_{it}, \qquad (20)$$

where $Abs(x)$ denotes the absolute value of the number x, and all other variables are defined as before. $CONTROLS_{i,t-\tau}$ is a set of controls that includes among other things absolute values of macroeconomic forecast errors.

Our estimations investigate errors made in forecasts of primary deficits in the same year, for which the forecasting horizon is effectively one year (as the WEO in April of a given year uses data constructed at the beginning of the year).[26]

[25] We have also performed the same analysis using the WEO vintage from the subsequent year (t+1) to obtain the realized value for year *t*. The findings, reported in the Online Annex Table F.1, show that, although with similar signs, the results become weaker with this alternative approach. Importantly, we rely on the latest WEO vintage because it reflects the most up-to-date historical series for each country, whereas the disadvantage of the alternative construction is the use of data which are not yet fully ex-post and prone to further change in the ensuing years. Typically, it may take up to 4 years for statistical bureaus to officially validate the "final" values.

[26] Note that the WEO also provides forecasts over longer horizons. However, this paper focuses on near-term forecasts, as projections over longer horizons tend to be less accurate due to increasing economic uncertainty. We have nonetheless run regressions for *t+3* and *t+5* horizons; the results are available upon request. These suggest mixed findings: the effect of fiscal institutions on reducing forecast errors weakens over longer horizons, though their impact on forecast accuracy remains relatively robust.

V.1. Forecasting Errors

Table 4 reports the estimates for the primary deficit projection error regression, provided by Equation (19). In all regressions we include year- and country-fixed effects (not separately reported), the lagged forecasting error, the forecasting error in the lag of the change in the debt ratio, and the growth and inflation forecasting errors ("surprises"). The latter two variables are defined analogously to the primary deficit projection error. Again, and for the same reasons as earlier, we estimate the model using instrumental variables, with an instrument set fully analogous to before, now constructed on the basis of the partially new set of variables on the right-hand side. That is, the instruments are constructed again as weighted averages (based on log-distances between capitals) of the right-hand variables, which now include forecasting errors of explanatory variables.

The lag of the primary deficit surprise is always highly significant at the 1 percent level with a coefficient of 0.20 or 0.21. The lagged change in the debt ratio and the inflation surprise come in with statistically insignificant coefficients. Not surprisingly, a positive surprise in real growth causes the primary deficit outcome to fall relative to its projection—an effect that is highly significant. The fiscal rule suspension variable is not significant. Column (1) shows that the effect of stronger fiscal rules on the primary deficit projection error is negative and significant.[27] A shift from the 25th to the 75th percentile of the distribution of fiscal rule strength, which amounts to a shift of 0.35 on a 0–1 scale, is associated with a reduction in the projection error of 0.85 percentage point of GDP, implying a long-run reduction in the projection error of 1.06 percentage point of GDP. Column (2) shows that the presence of a fiscal council is associated with a fall in the projection error of 1.18 percentage point of GDP, an effect that is significant at the 10-percent level.[28] An increase in fiscal council strength is associated with a statistically significant and negative effect on the primary deficit projection error (Column (3)). A shift from the 25th to the 75th strength percentile for fiscal councils—i.e., an increase by 0.23 on a 0–1 scale—would be associated with a reduction of the projection error by 0.42 percentage point of GDP. Using the coefficient of the lagged dependent variable in that column, the implied long-run effect is equal to -0.54 percentage point of GDP. Turning to the interactions, we find that fiscal rules strength exerts a significantly negative effect on the projection error when interacted with a fiscal council dummy (Column (4)) and with fiscal council strength (Column (5)).[29] These estimates suggest that the combination of councils and rules has a reinforcing role in reducing deficit surprises.

In the presence of a fiscal council, an increase in the fiscal rule strength from the 25th to the 75th percentile is associated with a statistically significant decline of 0.89 percentage point of GDP in the primary deficit projection error. The introduction of a fiscal council in the presence of a fiscal rule of average strength produces a reduction in the forecast error of 0.43 percentage point of GDP, which is not significant though. Further, an increase in rule strength from the 25th to the 75th percentile in the presence of a fiscal council of average strength lowers the forecast error by a statistically insignificant 0.74 percentage point of GDP, while an increase in council strength

[27] This pattern is the same for the individual types of rules, in particular a debt rule and a budget balance rule. For brevity we do not report these results here.

[28] The negative effect of councils on primary deficit forecasting errors is also found by Metz et al. (2026).

[29] The combinations of the coefficient of rules strength, fiscal council, and their interaction term are not statistically significant.

from the 25th to 75th percentile in the presence of a fiscal rule of average strength reduces the forecast error by a mere 0.04 percentage point of GDP.

Table 4. Regressions of the Primary Deficit Projection Error on Rules, Councils, and their Interaction, 2000–2024

Dependent Variable: Primary Deficit Surprise	(1)	(2)	(3)	(4)	(5)
Lag Dependent Variable	0.21***	0.21***	0.21***	0.20***	0.20***
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Lag Δ Debt Surprise	-0.01	-0.01	-0.01	-0.01	-0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Real Growth Surprise	-0.13***	-0.13***	-0.13***	-0.10**	-0.10**
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Inflation Surprise	-0.02	-0.02	-0.02	-0.02	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Fiscal Rule Suspension Dummy	0.25	0.19	0.21	0.25	0.24
	(0.30)	(0.28)	(0.28)	(0.30)	(0.30)
Fiscal Rule Strength	-2.39**			5.26**	4.42**
	(1.13)			(2.40)	(2.18)
Fiscal Council Dummy		-1.18*		1.87***	
		(0.61)		(0.69)	
Fiscal Council Strength			-1.81**		2.75***
			(0.86)		(1.07)
Fiscal Rule Strength x Fiscal Council Dummy				-7.78***	
				(2.03)	
Fiscal Rule Strength x Fiscal Council Strength					-9.85***
					(2.44)
R-squared	0.09	0.09	0.09	0.08	0.08
F test statistics	11.80	11.75	11.66	13.17	12.88
F test p-value	0.00	0.00	0.00	0.00	0.00
KP LM test p-value	0.00	0.00	0.00	0.00	0.00
KP Wald F test statistics	40.39	62.39	47.79	24.52	30.62
Observations	1,922	1,922	1,922	1,922	1,922
Number of countries	90	90	90	90	90
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Country fixed effect	Yes	Yes	Yes	Yes	Yes

Sources: Authors' estimations. See Annex A for all data sources used in the estimations.
Note: Kleibergen-Paap (2006) rk LM, Hansen J, and Kleibergen-Paap (2006) rk Wald F-statistics included. Standard errors are clustered at the country level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

We conclude this section with some variations on the baseline estimates reported in Table 4. First, we perform a robustness exercise that includes again the change in government bond spreads (see Online Annex Table F.2). The results show that the coefficients for the change in government spreads are statistically significant and positive. Our fiscal institutional variables retain statistically significant and negative coefficients.

Similar findings are obtained when we replace the change in government bond spreads with the dummy variable for years of election (Online Annex F). As Online Table F.3 conveys, the fiscal institutional variables continue to be statistically significant and negative. Once more, the dummy variable for electoral years is not statistically

significant in any of the regressions. Regarding the second political variable investigated, the number of changes in government (Online Annex Table F.4), all fiscal institutional variables lose significance except for the fiscal rule strength. The political variable itself is also not significant in those estimations.

A third test includes the World Uncertainty Index (WUI) in the estimations, given that an increase in uncertainty also makes it more difficult to project outcomes. The estimates are reported in Online Annex Table F.5. However, the WUI tends to be insignificant in all specifications.[30] The estimated association of the primary deficit projection error with the fiscal institution variables is broadly unchanged, although it is a bit weaker for the regressions with the interaction terms.

At last, we split the primary deficit error into a revenue error and a primary expenditure error (see Online Annexes G and H, respectively). We find that the presence and strength of a fiscal council exert a positive and significant effect on the revenue error, while their interactions with fiscal rule strength are negative and significant. The primary expenditure errors tend to decrease in the presence of strong fiscal rules and strong councils.

V.2. Forecasting Accuracy

We turn now to the determinants of forecasting accuracy—i.e., the absolute values of the forecasting errors of the primary deficit. The set of controls in Equation (20) includes the absolute values of the growth and inflation projection errors (surprises). Instrumentation is analogous to before.

Table 5 reports the results. The first lag of the dependent is highly significant, with a coefficient of 0.23 or 0.24. Hence, the degree of accuracy is correlated over time. The lagged change in the absolute value of the debt surprise also has a positive and highly significant coefficient. The other controls, except for the fiscal suspension dummy, are not statistically significant. The fiscal suspension dummy exerts a positive effect on the absolute value of the primary deficit projection errors, suggesting that periods in which rules are not active are associated with lower forecasting accuracy. When entered as “standalone”, the fiscal council dummy and fiscal council strength exert a positive and significant effect on the absolute value of the primary deficit projection errors (Columns (2)–(3)), suggesting that forecasting accuracy deteriorates. In the more fully-fledged specification with interactions, the effect on accuracy is less clear-cut. An increase in fiscal rule strength reduces it, while its interaction with the fiscal council dummy or fiscal council strength raises it. On net, an interquartile increase in rule strength lowers the average absolute value of the primary deficit projection errors by 0.86 percentage points of GDP (Column 4), while at the average council strength of 0.66 the effect of an interquartile increase in rule strength lowers the average absolute value of the primary deficit projection errors by 1.07 percentage points of GDP (Column 5).

This section concludes by discussing the tests that include additional controls on the estimations of forecast accuracy. Online Annex I reports the results of those tests. In many estimations, the institutional variables lose significance, indicating a lower robustness of the fiscal institutional variables in affecting forecast accuracy.

[30] The statistical insignificance of the WUI may be a bit surprising, as we would expect higher fundamental global uncertainty to feed into less accurate predictions of the eventual outcome of the primary deficit. However, the effects of more uncertainty on the fiscal realisations may go into both directions.

Regarding forecast accuracy of revenues and primary expenditure, Online Annexes J and K report the results for those two dependent variables, respectively. For revenue projections, we find that the presence of a fiscal council or its strength is associated with an increase in forecast accuracy, while their interactions with rule strength weaken accuracy (Online Annex Table J.1). Rules and fiscal council strength are associated with improvements in the accuracy of the primary expenditure projections (see Online Annex Table K.1).

Table 5. Regressions of the Accuracy of Forecasting Primary Deficits on Rules, Councils and their Interaction, 2000–2024

Dependent Variable: Absolute Value of Primary Deficit Surprise	(1)	(2)	(3)	(4)	(5)
Lag Dependent Variable	0.24***	0.24***	0.24***	0.23***	0.23***
	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Lag Δ Absolute Value of Debt Surprise	0.02***	0.02***	0.02***	0.02**	0.02**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Absolute Value of Real Growth Surprise	0.08	0.07	0.07	0.08	0.08
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
Absolute Value of Inflation Surprise	-0.01	-0.01	-0.01	-0.01	-0.01
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Fiscal Rule Suspension Dummy	0.58***	0.42**	0.42**	0.57***	0.60***
	(0.21)	(0.21)	(0.21)	(0.22)	(0.21)
Fiscal Rule Strength	-0.26			-7.55***	-7.16***
	(0.78)			(2.17)	(1.98)
Fiscal Council Dummy		0.74*		0.26	
		(0.42)		(0.48)	
Fiscal Council Strength			1.06*		0.49
			(0.58)		(0.76)
Fiscal Rule Strength x Fiscal Council Dummy				5.12***	
				(1.89)	
Fiscal Rule Strength x Fiscal Council Strength					6.25***
					(2.43)
R-squared	0.13	0.13	0.13	0.12	0.12
F test statistics	9.84	9.60	9.57	7.75	7.64
F test p-value	0.00	0.00	0.00	0.00	0.00
KP LM test p-value	0.00	0.00	0.00	0.00	0.00
KP Wald F test statistics	54.03	60.47	40.01	24.30	28.73
Observations	1,922	1,922	1,922	1,922	1,922
Number of countries	90	90	90	90	90
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Country fixed effect	Yes	Yes	Yes	Yes	Yes

Sources: Authors' estimations. See Annex A for all data sources used in the estimations.
Note: Kleibergen-Paap (2006) rk LM, Hansen J, and Kleibergen-Paap (2006) rk Wald F-statistics included. Standard errors are clustered at the country level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Hansen J statistics are not reported when the number of clusters is insufficient to calculate a robust covariance matrix.

VI. Concluding Remarks

This paper offers a first empirical evaluation of the effectiveness of fiscal rules and independent fiscal councils in promoting fiscally sustainable public finances by reducing budget deficits, reducing fiscal forecast errors, and

improving fiscal forecast accuracy, using a large sample of 90 countries, including 32 AEs and 58 EMDEs, spanning from 2000 to 2024, based on the IMF's comprehensive and recently updated dataset on fiscal rules and councils (Alonso et al., 2025).

The analysis begins by documenting stylized facts, such as the tendency for fiscal councils to be introduced primarily after one or more fiscal rules are already in place. In terms of design of fiscal rules, our stylized facts also show that the most common rules set limits on the budget balance, closely followed by debt rules.

The paper contributes to a better understanding of the contributions of fiscal rules, fiscal councils, and their interactions to fiscal discipline. It develops a simple theoretical framework to explain some stylized facts on the roles of fiscal rules and fiscal councils. The framework incorporates a budget balance rule and a fiscal council enhancing transparency on the economic outlook. It explains why some countries choose to adopt a fiscal council alongside existing rules while others do not, and it generates testable hypotheses predicting that stronger fiscal rules and councils individually reduce deficits and complement each other in achieving fiscal consolidation.

These hypotheses are empirically examined using the updated 2025 IMF databases on fiscal rules and councils. The results indicate that the strength of both fiscal rules and councils individually tends to have a statistical association with lower primary deficits, although their interaction terms are insignificant, possibly due to limited sample size. In terms of economic significance, our estimates suggest that shifting from the 25th to the 75th percentile of the distribution of the fiscal rule strength would be associated on average with a reduction in the primary deficit of 1.2 percentage points of GDP. An interquartile increase in the strength of fiscal councils would be associated with a reduction in the primary deficit of 0.4 percentage point of GDP.

The analysis further explores the impact of these institutions on primary deficit forecasting errors, finding that both rule and council strength tend to contribute to reducing forecast overoptimism and improving accuracy. In particular, our analysis suggests that an interquartile shift in the strength of fiscal councils would be associated with an average reduction in the forecast error of the primary deficit of almost 0.42 percentage point of GDP, while at the average fiscal council strength, an interquartile shift in the fiscal rule strength is associated with a 0.7 percentage point of GDP reduction in the forecast error.

These results shed light on policy implications regarding the design of fiscal rules and councils. Empirical results point to the case that fiscal rules and fiscal councils contribute to fiscal discipline, but through different channels and with varying magnitudes. The strength of rules and councils matters more than their mere presence. At the same time, since they contribute to fiscal discipline differently, policymakers would thus need to strengthen both fiscal rules and fiscal councils in parallel rather than only adopting one without the other. This is what is indeed increasingly happening. For some emerging markets and developing economies with limited capacity or expertise to establish a formal fiscal council, other avenues of strengthening fiscal oversight will be essential.[31] Our results show that the strength of fiscal rules and councils tend to be associated with stronger fiscal discipline, smaller forecasting errors, and improved forecasting accuracy. Further, putting in place strong fiscal rules might increase

[31] For example, these other avenues may include parliamentary budget committees or auditor offices (see Acalin et al., 2025).

the political acceptability of a council, as the strong rules already reduce the scope for profligacy and instill a culture of fiscal discipline.

A potential caveat to the analysis, however, is the limited variation in institutional variables, particularly regarding fiscal councils. The institutional landscape does not change often over the sample period, but the practices of fiscal councils (such as how their reports are written) could vary over time. The potential endogeneity of fiscal rules and councils to fiscal performance is another important consideration, which we have attempted to address with careful choices of instrumental variables to mitigate a potential estimation bias. Such limitations require avoiding strong conclusions. While this study seeks to address some of these limitations, further enrichment of institutional information and empirical identification are warranted.

Accordingly, several avenues for future research arise from this work. Future studies could assess the impact of fiscal rules and councils on public debt dynamics, including stock-flow adjustments beyond the primary deficit, which often happen “under the radar”. Future work could also analyze other fiscal rules (such as expenditure rules); delve further into differences in the effects of rules and councils between types of revenues and expenditures; or test other instrumentation strategies to address the potential endogeneity of fiscal rules and councils. The role of political factors, and how they interact with rules and councils, and their design, would be another fruitful avenue for further research. Additionally, expanding the fiscal council database to capture greater institutional detail could enable more granular tests of the hypotheses presented here.

Annex A: Data and Descriptive Statistics

Table A.I. Data Sources

Variable	Source
Fiscal Rule, Debt Rule, Budget Balance Rule	IMF Fiscal Rules Dataset, 1985-2024
Fiscal Council	IMF Fiscal Councils Database, 2024
Fiscal Rule Strength Index, Debt Rule Strength Index, Budget Balance Rule Strength Index	Alonso et al. (2025)
Fiscal Council Strength Index	Alonso et al. (2025)
Real GDP growth	Gross domestic product, constant prices, national currency, percent change; WEO Database (April 2025 vintage)
Inflation rate	Consumer Prices, period average, percent change; WEO Database (April 2025 vintage)
Primary deficit	General government primary net lending/borrowing, percent of GDP; WEO Database (April 2025 vintage)
Forecast for real GDP growth	WEO Database (April 2000 – April 2024 vintages)
Forecast for inflation rate	WEO Database (April 2000 – April 2024 vintages)
Forecast for primary deficit	WEO Database (April 2000 – April 2024 vintages)
Distance between capitals	The GeoDist Database, CEPII
Suspension of Fiscal Rules, activation of escape clauses	IMF Fiscal Rules Dataset, 1985-2024

Table A.II. Country Coverage

Country	Fiscal Rule	Fiscal Council	Advanced Economy	Emerging Market and Developing Economy	Country	Fiscal Rule	Fiscal Council	Advanced Economy	Emerging Market and Developing Economy
Albania				✓	Jordan				✓
Argentina	✓			✓	Kazakhstan	✓			✓
Australia	✓	✓	✓		Kenya	✓	✓		✓
Austria	✓	✓	✓		Kyrgyz Republic				✓
Belarus				✓	Lao P.D.R.				✓
Belgium	✓	✓	✓		Latvia	✓	✓	✓	
Bolivia				✓	Lithuania	✓	✓	✓	
Brazil	✓	✓		✓	Luxembourg	✓	✓	✓	
Bulgaria	✓	✓		✓	Malaysia	✓			✓
Burundi	✓			✓	Mexico	✓	✓		✓
Cambodia	✓			✓	Morocco	✓			✓
Canada	✓	✓	✓		Mozambique				✓
Chile	✓	✓		✓	Namibia	✓			✓
China				✓	The Netherlands	✓	✓	✓	
Colombia	✓	✓		✓	New Zealand	✓		✓	
Costa Rica	✓	✓		✓	Norway	✓		✓	
Croatia	✓	✓	✓		Pakistan	✓			✓
Cyprus	✓	✓	✓		Panama	✓	✓		✓
Czech Republic	✓	✓	✓		Paraguay	✓			✓
Côte d'Ivoire	✓			✓	Poland	✓			✓
Denmark	✓	✓	✓		Portugal	✓	✓	✓	
Dominican Republic	✓			✓	Qatar				✓
Ecuador	✓			✓	Romania	✓	✓		✓
Egypt				✓	Russia	✓			✓
Estonia	✓	✓	✓		Saudi Arabia				✓
Ethiopia				✓	Senegal	✓			✓
Finland	✓	✓	✓		Slovak Republic	✓	✓	✓	
France	✓	✓	✓		Slovenia	✓	✓	✓	
Gambia, The				✓	South Africa		✓		✓
Georgia	✓	✓		✓	Spain	✓	✓	✓	
Germany	✓	✓	✓		Sudan				✓
Greece	✓	✓	✓		Sweden	✓	✓	✓	
Guinea				✓	Switzerland	✓		✓	
Guinea-Bissau	✓			✓	Tajikistan				✓
Haiti				✓	Thailand	✓			✓
Hungary	✓	✓		✓	Tunisia				✓
Iceland	✓	✓	✓		Türkiye				✓
India	✓			✓	Uganda	✓	✓		✓
Indonesia	✓			✓	Ukraine				✓
Iran	✓	✓		✓	United Arab Emirates				✓
Ireland	✓	✓	✓		United Kingdom	✓	✓	✓	
Israel	✓		✓		United States	✓	✓	✓	
Italy	✓	✓	✓		Uruguay	✓	✓		✓
Jamaica	✓	✓		✓	Uzbekistan				✓
Japan	✓		✓		Yemen				✓

Sources: IMF Fiscal Rules Database, 1985–2024; IMF Fiscal Councils Database, 2024.

Table A.III: Descriptive Statistics, 2000–2024

Variables	Unit	Mean	Median	Standard Deviation	25th Percentile	75th Percentile	Minimum	Maximum	Observ.[1]
Full sample									
Fiscal rule and council indicators									
Budget balance rule index	Binary variable	0.55	1.00	0.50	0.00	1.00	0.00	1.00	2,225
Budget balance rule strength index	Index number	0.24	0.07	0.29	0.00	0.43	0.00	0.88	2,225
Debt rule index	Binary variable	0.45	0.00	0.50	0.00	1.00	0.00	1.00	2,225
Debt rule strength index	Index number	0.21	0.00	0.29	0.00	0.41	0.00	0.88	2,225
Fiscal council index	Binary variable	0.27	0.00	0.45	0.00	1.00	0.00	1.00	2,225
Fiscal council strength index	Index number	0.18	0.00	0.31	0.00	0.43	0.00	0.92	2,225
Fiscal rule index	Binary variable	0.61	1.00	0.49	0.00	1.00	0.00	1.00	2,225
Fiscal rule strength index	Index number	0.19	0.08	0.24	0.00	0.30	0.00	0.80	2,225
Macroeconomic and geographical indicators									
Inflation rate	Percent	6.34	3.54	13.30	1.78	7.30	-8.24	359.09	2,225
Primary deficit	Percent of GDP	0.65	0.83	3.81	-1.19	2.73	-19.39	17.45	2,225
Real GDP growth	Percent	3.34	3.44	4.06	1.51	5.58	-28.76	28.08	2,225
One-year ahead forecast for inflation rate	Percent	5.95	3.53	11.37	2.00	6.77	-3.73	249.79	2,225
One-year ahead forecast for primary deficit	Percent of exp. GDP[2]	0.85	0.90	5.24	-0.99	2.87	-60.74	29.82	2,225
One-year ahead forecast for real GDP growth	Percent	2.99	3.15	3.19	1.70	4.90	-35.00	19.97	2,225
Distance between capitals	Kilometers	7,871.7	7,273.3	1,527.1	6,703.3	8,940.5	6,517.2	13,331.8	2,225

Sources: IMF World Economic Outlook Database, Acalin et al. (2025), Alonso et al. (2025); and authors' calculations.
Note: [1] Observ. = observations. [2] exp. = expected.

Annex B: Case B of the Theoretical Framework

In this case $(\bar{d}, d_L^g, c)$ is such that $\bar{d}$ is strictly dominated by d_L^g.

This assumption holds if the concavity in $u(.)$ is sufficiently mild and c is sufficiently low.

For the case of a bad state of the economy, define $\bar{c}$ as the cost such that:

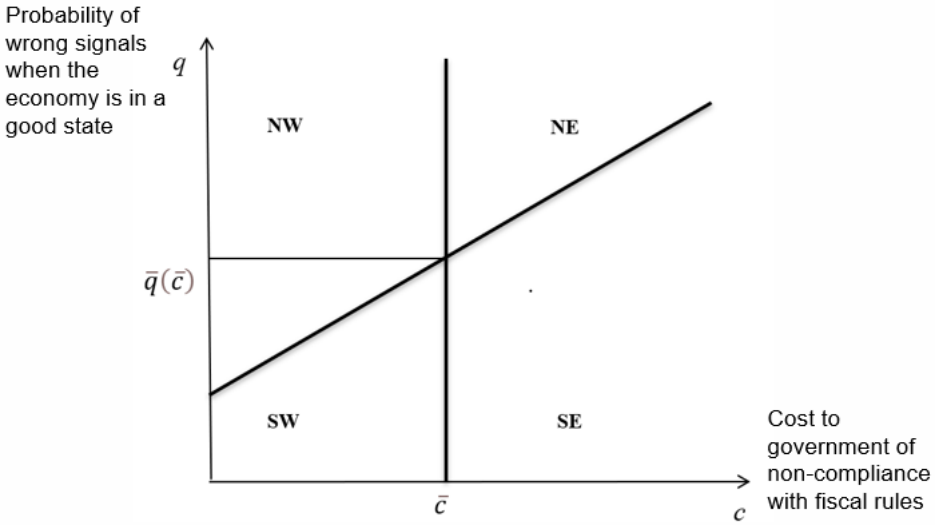
$$u(y_L + d_L^*) + \pi(\bar{y} - d_L^*) = u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c) \quad \text{(B.1)}$$

For the case of a good state of the economy, now define $\bar{q}(c)$ as the value of q such that:

$$u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c) = u(y_H + d_L^g) + \pi(\bar{y} - d_L^g - c) \quad \text{(B.2)}$$

We assume that $u(.)$ is such that there exists a combination $(c, \bar{q}(c))$ that makes both (B.1) and (B.2) hold. Because $u(y_H + d_L^g) > u(y_H + d_L^*)$, this is the case, for example, if π and c are sufficiently low. The function $\bar{q}(c)$ is decreasing in c.

Figure B.1 also defines four areas: (i) area North-West, where $LHS(B.1) < RHS(B.1)$ and $LHS(B.2) > RHS(B.2)$; (ii) area South-West (SW), where $LHS(B.1) < RHS(B.1)$ and $LHS(B.2) < RHS(B.2)$; (iii) area North-East (NE), where $LHS(B.1) > RHS(B.1)$ and $LHS(B.2) > RHS(B.2)$; and (iv) area South-East (SE), where $LHS(B.1) > RHS(B.1)$ and $LHS(B.2) < RHS(B.2)$.



Assume again that the economy starts in NW. The government's and legislature's utilities are, respectively:

$$\lambda\,[u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)] + (1-\lambda)[u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c)] \quad \text{(B.3)}$$

and

$$\lambda\,[u(y_L + d_L^g) + (\bar{y} - d_L^g)] + (1-\lambda)[u(y_H + d_L^*) + (\bar{y} - d_L^*)] \quad \text{(B.4)}$$

A reduction in q that keeps the system in NW lowers the government's expected utility but keeps the legislature's expected utility constant. At the switch into SW, the government's and legislature's utilities become, respectively:

$$\lambda\,[u(y_L + d_L^g) + \pi(\bar{y} - d_L^g - c)] + (1-\lambda)[u(y_H + d_L^g) + \pi(\bar{y} - d_L^g - c)] \qquad \text{(B.5)}$$

and

$$\lambda\,[u(y_L + d_L^g) + (\bar{y} - d_L^g)] + (1-\lambda)[u(y_H + d_L^g) + (\bar{y} - d_L^g)] \qquad \text{(B.6)}$$

Hence, at the transition legislature's expected utility jumps to a lower level, as in the good state of the economy the deficit rises even further from d_L^* to d_L^g, while the government's expected utility stays unchanged, because at the switching point $LHS(B.2) = RHS(B.2)$. Reducing q further does not affect the legislature's or government's utility. Overall, when the choice rests with the legislature, a sufficiently strong fiscal council will not be installed.

Now, start in NE. The government's and legislature's utilities are, respectively:

$$\lambda\,[u(y_L + d_L^*) + \pi(\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^*) + \pi(\bar{y} - d_L^* - (1-q)c)] \qquad \text{(B.7)}$$

and

$$\lambda\,[u(y_L + d_L^*) + (\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^*) + (\bar{y} - d_L^*)] \qquad \text{(B.8)}$$

A reduction in q while staying in NE lowers the government's expected utility but leaves society's expected utility unchanged. A sufficiently substantial reduction in q that brings the system into SE implies that the government's and legislature's utilities become, respectively:

$$\lambda\,[u(y_L + d_L^*) + \pi(\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^g) + \pi(\bar{y} - d_L^g - c)] \qquad \text{(B.9)}$$

and

$$\lambda\,[u(y_L + d_L^*) + (\bar{y} - d_L^*)] + (1-\lambda)[u(y_H + d_L^g) + (\bar{y} - d_L^g)] \qquad \text{(B.10)}$$

Again, at the switching point from NE to SE, the government's expected utility stays unchanged, because $LHS(B.2) = RHS(B.2)$. Since $d_L^g > d_L^* > d_H^*$, the legislature's expected utility is maximised at $d = d_H^*$ for $y_1 = y_H$, and $u(.)$ is strictly increasing and strictly concave, the legislature's expected utility jumps to a lower level when transitioning from NE to SE. A further reduction in q neither affects the government's expected utility, nor the legislature's expected utility. Overall, when the choice rests with the legislature, a sufficiently strong fiscal council will not be installed.

We notice that, because in none of the cases the government sets the deficit at $\bar{d}$, a tightening of the reference value $\bar{d}$ of the deficit rule does not affect the outcome of the deficit. Further, we notice that a sufficiently large reduction in q that brings the economy from NW to SW or from NE to SE generates an increase in the deficit from d_L^* to d_L^g in the good state but leaves the deficit in the bad state unchanged.

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